Guardian of Indian Skies

DID YOU KNOW THAT THE BALAKOT AIR STRIKE had a woman scientist in a pivotal role? The operation, which India carried out across the border as retaliation for the Pulwama terror attack in 2019, employed 12 Mirage 2000 aircraft. To evade detection by Pakistani radars, most Mirage 2000 jets flew at low altitudes, with India's Airborne Warning and Control System (AWACS), also known as the 'Eye in the Sky,' providing radar cover to the fighter jets. AWACS's architect, Dr Suma Varughese, closely monitored the movement of the 'bird' from a classified location until it returned to base.

"It was the most exhilarating experience for me. What was conceptualised 10 years ago on the drawing board coming through the sky was the ultimate happiness I could wish for," Suma says with a sense of pride. The same 'bird' hovered over the Delhi sky during the G20 Summit in September to keep an eagle eye out for any unwanted flying objects.

The AWACS, crafted as a force multiplier for the Indian Air Force, was engineered by the Centre for Air Borne Systems of the Defence Research and Development Organisation (DRDO). The system consists of multiple sensors for surveillance and signals intelligence, helping in air defence operations.

India joined the elite group of five nations possessing the capability to deploy a home-made early warning aircraft, when AWACS soared over the Rajpath during the Republic Day parade in 2017.

Bengaluru-born Suma obtained her master's in Computer Science from Mysore University and then another master's at the Indian Institute of Science, Bangalore, specialising in radar antenna measurement systems. Her professional journey began at the DRDO's Electronics and Radar Development Establishment. "Curiosity, wonderment, passion and learning to challenge are the key motivations that drove me into the complex defence research field."

ACCOMPLISHMENT

Created the Airborne Warning and Control System or AWACS, commonly known as 'Eye in the Sky', which provides the radar cover to fighter jets

The visibility of a battlefield—how early you can detect a target—is crucial, she says. "AWACS gives you that vantage point to track your enemy targets. And today with evolving technologies, we can maintain radio silence for a long time, which makes you deceptive." India is safe as long as Suma has her eye in the sky. ■ ***–Pradip R. Sagar***

PARITY PRESCRIPTION

"I don't think men are better than women. A woman just learns to assert herself. I had never aspired that I would be here"

GRATEFUL TO... DRDO ex-head Dr V.S. Arunachalam, who started a computer science course in the early 1980s. It paved the way for her

Suma Varughese, ***58***

Director-General, Micro-electronic Devices, Computational Systems and Cyber Systems, DRDO



CHANDRADEEP KUMAR